

Weeks 2-10

Extra / grades 6-7

One optional investigation per week; offer by readiness.

Grade bands are entry points. Begin with objects, then try to explain what happens.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Keep facilitator solutions separate from student pages.

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Print US Letter, single-sided, at 100% / Actual Size.

Page ranges above count this cover as page 1. Bookmarks open each week.

Weekly footers retain their own page numbers and activity IDs for the use log.

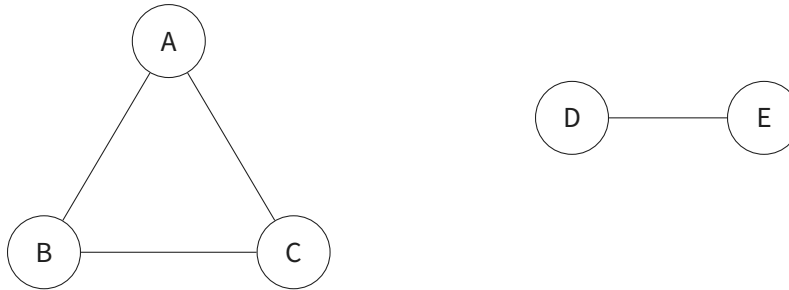
Revised September 20, 2026. Planned activities; record actual use after teaching.

A theorem for any network

Name: _____

Date: _____

Start every lamp OFF. One move flips the two ends of one drawn edge. A target is a chosen set of lamps to leave ON.



1. Is even enough?

Can you light only A and D? Only A, C, D, E? Give a move list for one and an obstruction for the other.

My results and reasons:

2. Repair the conjecture.

What must be true of the number of target lamps in **each connected piece** of a network? Prove necessity. Then prove sufficiency by pairing target lamps and following paths. If paths overlap, what happens to a shared edge?

An if-and-only-if rule for every finite network:

3. Find the cheapest repair on a tree.

Draw a connected network with no closed loops (a **tree**), using at least six dots. Choose an even number of target lamps. Removing an edge splits the tree in two. Prove that an edge is pressed an **odd number of times** exactly when one side has an odd number of targets. Why is the solution using each edge at most once **unique and shortest**?

A labeled example and an argument:

Try it with objects. Explain by pointing, talking, or drawing.

The slowest eight-letter machine

Name: _____

Date: _____

Choose a permutation of A–H: every letter appears once in each row of a substitution key. Repeat it. Maximize the first positive turn when **all eight letters** are back. Use cards and arrow loops.

1. Beat one big loop.

An eight-letter loop takes eight turns. Can two or more loops take longer? Build a key and show why its claimed return is the *first*.

My key or loops, and its return time:

2. A complete upper bound.

The return time is the least common multiple of the loop sizes. Sort by the **largest loop**. Complete the cases below; explain why no partition of eight was missed.

Largest loop	Possible remaining loops	Best return time
8	none	
7	1	
6	2 or 1+1	
5	partitions of 3	
4	partitions of 4	
At most 3	only 1, 2, 3	

3. A theorem and a next question.

Use your construction and bound to prove the optimum. Then predict the optimum for nine letters. Must adding one letter make the optimum *strictly* bigger?

My theorem, proof, and nine-letter prediction:

Try it with objects. Explain by pointing, talking, or drawing.

A perfect shuffle has a clock

Name: _____

Date: _____

Use eight cards 0–7 in a row, leftmost = top. Cut into two equal halves **without reversing either half**. Interleave, starting with the first half: 0 1 2 3 / 4 5 6 7 becomes **0 4 1 5 2 6 3 7**. This is an out-shuffle.

1. Shuffle until home.

Repeat the exact rule. How many shuffles first restore the whole original order? Record rows or track each card's *position*.

My rows or position cycles:

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2. Explain the rule with remainders.

Positions are numbered 0–7. Track where a card at old position x goes. For $x=0$ through 6, test “double x , then take the remainder after division by 7.” What happens to position 7?

Old position	0	1	2	3	4	5	6	7
New position								

3. Prove the return time.

After t shuffles, a position below 7 becomes the remainder of $2^t x$ after division by 7. Find the first t with 2^t leaving remainder 1. Why does checking $x=1$ show that no earlier return works?

My proof, including the fixed last card:

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Go further: Predict the first return for 6 cards and for 16 cards. For $2m$ cards the relevant modulus is $2m - 1$. Why does twice as many cards not necessarily mean twice as many shuffles? **Research bridge:** Diaconis, Graham, and Kantor studied which permutations combinations of perfect in- and out-shuffles generate. This page proves the single out-shuffle return formula in a small case.

Try it with objects. Explain by pointing, talking, or drawing.

Small swaps force a lower bound

Name: _____

Date: _____

New clue type: an entry clue fixes the height **in one cell**. There are no outside visibility clues. Rows and columns still contain 1–4 exactly once.

1. A switch that preserves every rule.

Copy this Latin square. In the upper-left 2-by-2 block, exchange the 1s and 2s. Why is the new square still legal?

1	2	3	4
2	1	4	3
3	4	1	2
4	3	2	1

2. Four independent hiding places.

Each bold 2-by-2 block has a similar switch. Suppose you reveal at most three entry clues from the original square. Show that a second completion always remains.

Why must one whole switch escape the clues?

3. Is touching every block sufficient?

Reveal just cells (row 1, column 1), (1,3), (3,1), (3,3). They touch all four bold blocks. Can you still make another square?

An alternative completion or a convincing reason:

Go further: A *Latin trade* changes some entries and preserves every row and column. Explain why a uniquely determining entry set must touch every possible trade, not merely these four.

Build first. Explain by pointing, talking, or drawing.

Five places, only four tests

Name: _____

Date: _____

Write R as 0 and B as 1. A score still counts correct **positions**. Choose all tests in advance: 00000, 00011, 00101, 01001.

1. Decode a real example.

These four tests score 2, 2, 2, 0 in that order. Find the secret. Verify every score.

Secret and check:

2. Why could this always work?

Let the secret be $x_1x_2x_3x_4x_5$, with each x_i equal to 0 or 1. The first score tells the total number of 1s. What does changing two test positions from 0 to 1 tell you?

Relate the score changes to these sums: $x_4 + x_5$, $x_3 + x_5$, $x_2 + x_5$.

3. The only tricky case.

A pair-sum of 0 or 2 forces both bits. What if all three pair-sums are 1? Try $x_5 = 0$ and $x_5 = 1$. Can they have the same total number of 1s, even after you choose x_1 ?

Why the four answers identify every possible secret:

A geometric translation: Join binary words that differ in one position. Distance is the number of mismatches, so distance = 5 minus score. These four test words are landmarks that distinguish all 32 vertices of a five-dimensional cube.

Build first. Explain by pointing, talking, or drawing.

Two winning piles can make a loss

Name: _____

Date: _____

Use moves 1, 3, 4. Now there are **two piles**; a turn changes exactly one pile. The player who takes the very last counter wins.

1. W and L forget too much.

Play from piles (1,1), then (1,4). Each individual pile is W. Are both combined games W? Explain your best play.

2. Give each pile a richer label.

Let $g(0) = 0$. For a larger pile, list the g -values reachable in one move; give the pile the **smallest nonnegative integer missing** from that list. For example, $g(1) = 1$ and $g(2) = 0$.

Pile	0	1	2	3	4	5	6	7	8	9	10
g	0	1	0								

3. Test an equality rule.

Conjecture: two piles are losing together exactly when their g -values are equal. Check (1,3), (4,6), and (5,6). Find a winning move wherever possible.

Checks and moves:

4. Prove the rule for two piles.

Why can a move never keep equal labels equal? If labels differ, why can you move from the larger label to the smaller one? Use the “smallest missing” definition.

The two parts of my proof:

Beyond this page: Three piles require binary XOR, not ordinary addition or pairwise equality. These labels are Sprague–Grundy values.

Build first. Explain by pointing, talking, or drawing.

One new move, a new world

Name: _____

Date: _____

Use TWO piles. You may remove any positive number from one pile, OR remove the SAME positive number from both. Take the last counter to win. This is Wythoff's game.

1. Break the matching strategy.

Why is $(4, 4)$ no longer a trap? Classify all starts (a, b) with $0 \leq a \leq b \leq 7$. Work in order of $a + b$; a trap has no move to an earlier trap. Treat $(0, 0)$ as terminal.

2. Build a smaller list.

Start with $(0, 0)$. Repeatedly choose the smallest nonnegative integer not yet in either column as a_n , then put $b_n = a_n + n$.

n	0	1	2	3	4	5
a_n	0	1				
b_n	0	2				

Does this list match your small-game traps? Explain why no rook move and no equal-take move can connect two listed pairs, even when pile order is swapped.

What is proved? What still needs to be proved?

3. An unexpected number.

Let $\phi = (1 + \sqrt{5})/2 \approx 1.618$. A known theorem says the pairs are $(\lfloor n\phi \rfloor, \lfloor n\phi^2 \rfloor)$, together with swapped pairs. Floors mean round down. Test $n = 1, \dots, 5$. Why does $\phi^2 = \phi + 1$ explain the differences $b_n - a_n = n$?

Harder: To prove the list is the whole losing set, what must you show about a position *outside* it? Try finding such moves from $(2, 4)$, $(5, 8)$, and $(8, 12)$.

Try it with objects. Explain by pointing, talking, or drawing.

Change the slope

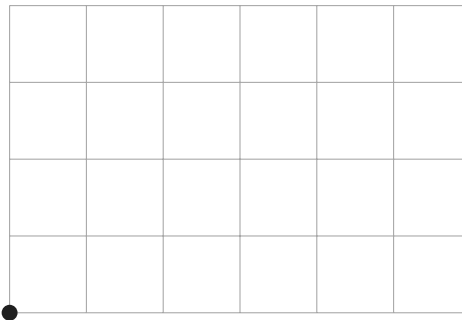
Name: _____

Date: _____

Use a UNIT SQUARE. Start at $(0, 0)$ heading up-right with slope $s = \text{vertical distance} / \text{horizontal distance}$. Unfold reflections into the straight line $y = sx$. Stop at a corner.

1. Rational directions.

Trace unfolded slopes $2/3$ and $3/2$. If $s = p/q$ with positive coprime integers p, q , why is the first lattice point (q, p) ? Predict the ending corner and count bounces.



each little square is a whole reflected table

2. A path that never reaches a corner.

Take $s = \sqrt{2}$. If the unfolded path reached an integer point (u, v) with $u > 0$, what would v/u equal? Why is that impossible?

A proof, not just a long drawing:

3. Never hitting is not the same as going everywhere.

Start instead at $(0, 1/2)$ with slope 1. Trace four bounces. Does this path hit a corner? Does it explore the whole square?

A theorem beyond this page: The $\sqrt{2}$ -slope path launched from $(0, 0)$ above is dense: it comes arbitrarily close to every point. Your no-corner argument alone does *not* prove this. Unfolding leads to straight-line motion on a torus, a subject in dynamical systems.

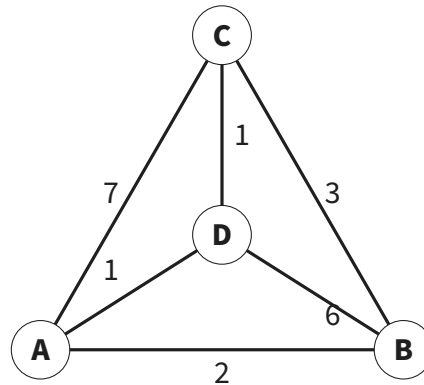
Try it with objects. Explain by pointing, talking, or drawing.

Cheapest pair first?

Name: _____

Date: _____

Start at A, travel every road at least once, return to A. Numbers are road lengths. Find the minimum total length. Repeated trips count again.



1. Roads and shortest connections differ.

What is the shortest connection from A to C? From B to D? A connection may use several roads.

2. Compare all three pairings.

All four vertices are odd. Pair them and repeat a shortest connection for each pair. Compare $AB+CD$, $AC+BD$, and $AD+BC$. Does beginning with the cheap pair AD guarantee the best total?

Connection costs, best pairing, and predicted total:

3. Certify the minimum.

Give an actual closed route attaining your total. For a lower bound, explain why the extra road copies in *any* closed solution can be separated into paths pairing the four odd vertices, plus possible closed loops.

Beyond four vertices: With $2k$ odd vertices, there are $(2k-1)(2k-3)\cdots 1$ pairings. Research on the Chinese postman problem replaces trying them all with efficient minimum-weight matching algorithms.

Try it with objects. Explain by pointing, talking, or drawing.